

Approaches to handle data hazard

- Schedule: Programmer explicitly avoids scheduling instⁿs that would create data hazard.
- Stall: Hardware includes control logic that freezes earlier stages until preceding instⁿ has finished producing data value.
- Bypass: Hardware data path allows data to be sent to an earlier stage before preceding instⁿ has left the pipeline.
- Speculate: Guess that there is not a problem, if incorrect kill speculative Instⁿ & restart.

Control Hazard

<u>ADD</u>	F	D	X	M	W
<u>ADD</u>			F	D	
<u>Sub</u>				F	

1 2 3 4 5

I1	096	ADD	F	D	X	M	W
I2	100	B 304		F	D	X	M
I3	104	ADD			F	nop	
I4	<u>304</u>	ADD				F	